Korea Technology and Information Promotion Agency for SMEs

*(max 10 pages)*

# Technology Proposal for 2026 TIPA* International Joint R&D Program

**(TIPA: Korea Technology and Information Promotion Agency for SMEs)**

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

## 1. Technology Proposer Information

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|---|---|---|---|---|
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## 2. Technology Overview

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|---|---|---|---|---|
| Title of Proposed Technology | Connection module to connect fiber reinforced plastics and metal constructions for load transfer and electrical contact | | | |
| Korea's 12 National Strategic Technology Fields | Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility (x), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine (x), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (x), Quantum ( ) | | | |
| Industrial Technology Classification [1] | **Technology Field 1** (Large Category) | | **Technology Field 2** (Large Category) | |
| | *Mechanical Materials* | | *Chemicals* | |
| | Medium Category | Small Category | Medium Category | Small Category |
| | *Automobile / rail Vehicles* | *Vehicle Body and Light-weighting Technologies* | *Fiber Products* | *Other Fiber Product Technologies* |
| Concept of Proposed Technology | *Describe the core concept, main functions, and potential uses of the proposed technology*<br>The aim is to generate a hybrid insert for the integration into fiber reinforced plastics applications. The insert consists of fibers that are locally covered with metal structures. The integrated fiber stays fully intact. The integrated fibers can be directly bonded to the surrounded fiber reinforced plastics structure. The metal point can be used as connection point to metal structures using different joining technics in laminated state like welding, press fit, screwing, glueing.<br>Finally potential use cases can be addressed in automobile and rail vehicles application where metal needs to be connected to fiber reinforced plastics or change steel housing parts to fiber reinforced plastic using the same metallic core structure. It is also possible to generate innovative battery or electronic housings for more flexible battery change concepts for electric vehicles. | | | |

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| **Rationale for Joint R&D Project** | *Describe the need for this joint R&D project and why collaboration is required*<br>Since the research institute developed a unique method to build up the hybrid inserts, the integration into real applications needs to be discussed with the industrial partner. Also, the final textile integration process must be covered by the industrial partner. During the interaction, also potentials for new design concepts need to be discussed. This can be done only in a direct collaboration in a joint R&D project. |
| **Key Words** | 1. *Provide key terms that describe the proposed technology and its main features (min 3, max 5)*<br>2. Fiber reinforced plastic<br>3. Hybrid insert to transfer load<br>4. Hybrid insert for better electrical connection<br>5. Replacement of metal sheets |

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## 3. Research and Development Objectives & Plan

| Technology Readiness Level (TRL)[2] | Start of the Project | Level (4) | End of the Project | Level (6) |
|---|---|---|---|---|
| R&D Objectives | *Describe the specific technologies you aim to develop through this R&D project*<br>Starting point is the discussion with the industrial partner about the aimed application and opportunities to increase the performance with the supposed technology of hybrid insert. Then a specific hybrid insert needs to be defined and designed. The specific sintering strategy needs to be developed and the final integration in the fiber reinforced plastic should be agreed. | | | |
| R&D Target Specifications | *Describe the target performance specifications to be achieved through this R&D project, including quantitative values such as measurable parameters*<br>- Replacement of metal sheets with fiber reinforced plastic using conventional joining techniques like welding, screwing<br>- Reduce the weight of sheet element by 20%<br>- In case of mechanical overload: Introduce a plastic deformation in the connection to lower the risk of failure of the fiber reinforced plastic structure<br>- | | | |
| R&D Plan | *Provide detailed plan outlining the research approach, key tasks, methodologies, expected milestones, and the timeline required to achieve the stated R&D objectives*<br><br>Phase 1 (6 months): Evaluation of actual design and opportunities of integration<br>Korea: Produce actual design of fiber reinforced plastic sheet or structures with actual connection method and mechanical evaluation data, provide load scenario<br>Germany: Development of possible integration opportunities of hybrid insert for the specific load scenario<br><br>Phase 2.1 (20 months): Integration of hybrid insert and evaluation in actual design<br>Korea: Provide actual design, parts, components for integration process of hybrid insert; Mechanical evaluation of integrated version and comparison with original setup.<br>Germany: Development and manufacturing of hybrid insert; Integration into model samples to characterize mechanical behavior; supply to Korea partner of hybrid insert for integration in final part.<br><br>Phase 2.2 (10 months): Adaption and integration of hybrid insert in new design and build up of prototype, evaluation<br>Korea: Development and manufacturing of new design with integrated hybrid insert for mechanical and further evaluation<br>Germany: Development and manufacturing of hybrid insert and integration in new design models; mechanical evaluation<br><br>Milestone (month 22): Integration and mechanical evaluation document better mechanical performance of the integrated setup compared to the original setup. | | | |

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## 4. R&R (Roles and Responsibilities)

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| **Expected R&R of Research Institute** | *Describe the expected roles and responsibilities of the research institute, such as technical contributions, research scope, resources to be provided, and specific tasks to be provided, etc.*<br>- Development and manufacturing of hybrid insert with powder metallurgical methods<br>- Integration of hybrid insert into actual and new design models<br>- Mechanical evaluation of actual and new design models<br>- Further material characterization |
| **Expected R&R of Korean SME Partner** | *Describe the expected roles and responsibilities of Korean SME, such as technical contributions, development tasks, testing/validation activities, resource commitments, and collaborations efforts, etc.*<br>- Provide actual design of fiber reinforced plastic sheet or structure with conventional connection<br>- Perform mechanical evaluation<br>- Provide, manufacture parts and components to build up a whole system with and without the integration of hybrid insert<br>- Develop new designs with integrated hybrid insert<br>- Manufacturing new design as prototype<br>- Perform different connection trails |

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## 5. Expected Outcomes/Impact

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| **Expected Outcomes and Impact** | *Explain the expected outcomes of this joint R&D project, including technological achievements, commercialization potential, and broader industrial or economic impact, etc.*<br>- After integration of hybrid insert the connection between metal structure and fiber reinforced plastic is stronger compared to conventional one.<br>- Including the hybrid insert in the design phase of new products will generate an unique selling point for the industry (connection to different metals like titanium, copper, aluminum, steel are possible)<br>- Integration of hybrid insert gives the opportunity to handle a fiber reinforced plastic sheet like a metal sheet in term of variety of bonding possibilities<br>- Increase market penetration opportunities through increased mechanical performance compared with weight saving |

## 6. References

(1)

(2)

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## ☐ Terms:

### 1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

### 2) Technology Readiness Level (TRL)

| Description | | |
|---|---|---|
| ·A consistent and objective indicator of the technological maturity of key element technologies | | |
| | **Basic Research Stage (1-2)** | 1st Stage: Basic theories and experiments<br>2nd Stage: Proof of Concept, such as practical ideas or patents |
| | **Experimental Stage (3-4)** | 3rd Stage: Verification of basic performance at a laboratory scale<br>4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale |
| | **Prototype Stage (5-6)** | 5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems<br>6th Stage: Production and performance evaluation of prototypes at a pilot scale |
| | **Productization Stage (7-8)** | 7th Stage: Reliability testing and evaluation by potential customers<br>8th Stage: Prototype certification and standardization |
| | **Commercialization Stage (9)** | 9th Stage: Commercialization |